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LATENT PROBABILITY MODELS WITH DIRECT EFFECTS BETWEEN INDICATORS

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1. Introduction

Social scientists are well aware of the gap that often exists between theoretical concepts and operational definitions, and have tried to bridge this gap by the construction of formal models by which one might judge the consequences of various forms of unreliability and invalidity.

Most of this work has been done in the area of causal analysis. Through path analytic models in which a complex mixture of both measured and unmeasured variables appear, a deeper insight into problems of reliability and validity has been achieved [1].

The usefulness of these models however is limited by the fact that the various forms of path analysis require that the variables are measured on at least interval level, and assume that all relations are linear. However, there are models with latent (unmeasured) and manifest (measured) variables that are capable of handling data on a nominal level of measurement. The foundations of these models have been laid by Lazarsfeld and Wiggins [2]. These so-called latent structure or latent probability models have one critical assumption that will be discussed in this article. In applying a latent structure to the data one has to assume that the relations between the manifest variables, the indicators, are fully explained by the relations between the latent variables among each other and the relations between the latent and the manifest variables. This so-called assumption of local independence implies therefore that there are no direct causal links between the indicators.

In a number of situations this could be a very improbable hypothesis. If, for instance, a respondent is asked twice to express his opinion on the same subject within the same interview or within a short period of time, the chances are that his answer on the first occasion will influence

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the direction of his answer on the second occasion, regardless of his true opinions in this matter.

Every time a test retest effect is assumed, these models cannot be applied. It is the purpose of this article to show how the strategy underlying the latent probability models can be changed to account for these test retest effects. In the next Section the basic assumptions of the latent probability models will first be demonstrated by presenting a very simple model borrowed from Wiggins (1973). In Section 3 this simple model will be changed to incorporate possible direct effects between the indicators. Readers already familiar with the work of Lazarsfeld and Wiggins might want to skip Section 2 and go directly to this section. Finally, a more elaborate model is presented which makes use of a general solution for latent probability models developed by Goodman (1974a, 1974b).

2. Latent Probability Models: an example

Six months before the 1971 general elections of the Dutch Parliament a nationwide panel study was started, in which the respondents were interviewed each month about their position on a number of political issues [3]. The answers of a subsample of the respondents to the question whether they would vote at the next elections, are given in Table I (first two waves). If one accepts the data of Table I as shown, a conclusion might be that vote intention is not a stable attribute. 16.5% of the respondents changed their minds from November to December, thereby increasing the percentage of potential voters from 83.0% to 89.2%. Further analyses could be done by using data from more time points to test e.g. whether these changes are congruent with a Markov

TABLE I

Proportion of Voters and Nonvoters in November and December

Vote Intention in November (Variable A)	Vote Intention in December (Variable B)		
	voter		total
	voter	nonvoter	
	.779 (652)	.051 (43)	.830 (695)
	.114 (95)	.056 (47)	.170 (142)
total	.892 (747)	.108 (90)	1.000 (837)

model. Essential to such an approach is the assumption that the manifest changes are real, indicating true changes in the respondents position.

It is more likely however that these data are not perfect indicators of the true vote intention of the respondent, but are contaminated with error. Surely, some mistakes will be made by the interviewer; a few errors during the data processing are unavoidable; also, the mood the respondent happens to be in, gives his answer partly the appearance of a random event. Because of all these and related factors the answers of the respondents do not perfectly reflect their true attitude, and so the manifest changes are not necessarily indicative of the real changes that take place.

Latent probability models try to cope with this rather complex situation. An example of this approach applied to Table I might go as follows. There exists a latent variable vote intention with two (latent) classes, voter and nonvoter, that reflect the true position of the respondent. Every respondent is in one and only one latent class: he is either a voter or a nonvoter. This latent variable is assumed to be stable: there are no latent changes from voter to nonvoter or vice versa. Now the manifest variables *A* and *B*, the answers on the question of vote intention in November and December, are not perfect measures of the true, latent positions of the respondents. Rather given his latent position, a respondent has a certain *probability* of classifying himself as a voter or a nonvoter. This probability is assumed to be the same for all the respondents in the same latent class (assumption of homogeneity). Furthermore, one assumes that within a certain latent class the answers

TABLE II

The parameters of a Latent Probability Model with changing Latent Probabilities and no Latent Change; two Manifest Variables *

Latent variable	Relative frequencies	Probability of answering	
		Var. A	Var. B
		1.	2.
Class I	v_I	a	$1 - a$
Class II	v_{II} ($= 1 - v_I$)	$1 - a$	a
		b	$1 - b$

* v_I = relative frequency of latent class I. v_{II} = relative frequency of latent class II. a = probability of answering "correctly" on variable A. b = probability of answering "correctly" on variable B.

to both questions are independent. So this so-called assumption of local independence implies that for a given person the probability of answering "voter" in December is, for instance, .80, whatever his answer was in November. Or to put it in a different language: while holding the latent variable constant, there will be no correlation between the variables *A* and *B*.

With these assumptions and the parameters mentioned in Table II it is possible to compute the proportions of the manifest variables expected under this model. As can be seen from Table II, one more assumption has been made: the probability of a "correct" answer corresponding with one's true, latent position, is the same for both latent classes.

Now the equations expressing the (expected) manifest proportions as a function of these parameters are as follows, where p_1^A = the proportion that answers 1 on variable *A*; p_1^B = the proportion that answers 1 on variable *B*; and p_{11}^{AB} = the proportion that answers 1 on *A* and *B*:

$$\begin{aligned} p_1^A &= av_1 + (1 - a)(1 - v_1) \\ p_1^B &= bv_1 + (1 - b)(1 - v_1) \\ p_{11}^{AB} &= abv_1 + (1 - a)(1 - b)(1 - v_1) \end{aligned}$$

By substituting the actual frequencies from Table I in the equations above (and so interpreting the score 1 as "voter") one obtains the following estimates for the parameters *a*, *b* and v_1 [4]:

$$a = .876 \quad b = .946 \quad v_1 = .940$$

Remembering the assumption of no latent change, one would conclude that 94% of the respondents are truly "voters".

The manifest changes in Table I are due to "unreliability". With regard to this "unreliability", it appears that the answers from the second interview were more "reliable" than from the first interview. The applicability of this model (and the above conclusions) may be tested by using data from more points in time (e.g. three interviews). Matters of estimating and testing latent probability models are extensively discussed in Lazarsfeld (1968), Wiggins (1973), and esp. Goodman (1974a).

This example should have made clear the basic elements of the latent probability approach. One postulates one or more latent variables that reflect the true positions of the respondents. Given this latent position, there is a probability of giving certain answers. Moreover, assumptions of homogeneity and local independence are made.

With these basic elements a rich variety of models can be (and has been) constructed [5]. One can use different numbers of indicators

(manifest variables); one can make a number of assumptions about the probabilities of giving certain answers; one can make allowances for all sorts of latent changes to take place. But, it has been said before, all these models assume that there are no direct (causal) links between the indicators. This assumption will be relaxed in the next section.

3. Some simple Latent Probability Models with Direct effects from one Indicator to another

To get a better idea of the consequences of relaxing the assumption of local independence, the same example as in the previous section will be used (Table I). A test retest effect will be assumed between the answer in November (variable *A*) and the answer in December (variable *B*).

Stated in terms of the latent probability models this means that given the latent position one is in, one has a different probability of scoring 1 (or 2) on variable *B* depending on the answer on variable *A*. A possible model along these lines is given in Table III. In this particular model the assumption is made again that there are no latent changes. Furthermore one assumes that there is a "basic" probability *a* of answering "correctly".

However for the second interview, this basic probability *a* will be increased or decreased by an amount *c* depending on the answer given on the first interview. So for instance, the probability of scoring "1" on variable *B* is $a + c$, given a person is in latent class I and has scored "1" on variable *A*, but this same probability will be $a - c$, if the respondent is in latent class I, but has given the answer "2" on variable *A*. The equations, expressing the (expected) frequencies, as a function of the

TABLE III
The parameters of a Latent Probability Model with Direct Effects between the Indicators

Latent variable	Relative frequencies	Probability of answering			
		<i>A</i> = 1		<i>A</i> = 2	
		Var. <i>A</i>	Var. <i>B</i>	Var. <i>B</i>	Var. <i>B</i>
		1.	2.	1.	2.
Class I	v_1	<i>a</i>	$1 - a$	$a + c$	$1 - a - c$
Class II	$1 - v_1$	$1 - a$	<i>a</i>	$1 - a + c$	$a - c$
				$1 - a - c$	$1 - a - c$
				$a - c$	$a + c$

parameters of this model, are as follows:

$$p_1^A = av_1 + (1 - a)(1 - v_1) \quad (1)$$

$$p_{11}^B = av_1(a + c) + (1 - a)v_1(a - c) + (1 - a)(1 - v_1)(1 - a + c) \quad (2)$$

$$+ a(1 - v_1)(1 - a - c) \quad (3)$$

$$p_{11}^{AB} = av_1(a + c) + (1 - a)(1 - v_1)(1 - a + c) \quad (4)$$

Eqn. (1) can be rewritten as:

$$v_1 = \frac{p_1^A + a - 1}{2a - 1} \quad (5)$$

Eqn. (2) is very different from the analogous equation of the previous model without test retest effects. Here p_1^B is the result of the sum of four groups, instead of two. These four groups are necessary because the probabilities of answering "1" or "2" on variable B not only depend upon the latent class one is in, but also on the answer on variable A . Substituting (4) into (2), and rearranging terms, one gets:

$$c = \frac{p_1^B - p_1^A}{2p_1^A - 1} \quad (6)$$

By substituting (4) into (3) one gets eventually:

$$p_{11}^{AB} = a^2 - a + p_1^A(c + 1). \quad (7)$$

Given that a probability of answering "correctly" of less than .50 does not make good sense, the solution to eqn. (6) is:

$$a = .50 + \sqrt{.25 - p_1^A(c + 1) + p_{11}^{AB}} \quad (8)$$

By using the manifest proportions of Table I, and eqns. (5), (7) and (4) the parameters are estimated as follows:

$$a = .847 \quad c = .094 \quad v_1 = .976$$

Comparing these results with the results from Section 2, one sees that the estimate for a is a little lower (.847 vs. .876) and accordingly the estimate for v_1 is somewhat higher (.976 vs. .940). But the main difference between these models is the appearance of the quantity c . It means in this case that the probability of answering "voter" (or "non-voter") during the second interview, is, for a given latent position, .188 ($= (a + c) - (a - c) = 2c = 2 \times .094$) higher if one has qualified oneself as a "voter" (or "nonvoter") before than if one has not done this.

Of course without additional data there is no way of telling whether this model with test retest effects fits the data better than a model without these effects or whether these assumptions really make sense.

TABLE IV

The Parameters of a Model with Direct Effects between Indicators - "Influences" from Variable $A = 1$ to Variable B

Latent variable	Relative frequencies	Probability of answering	
		$A = 1$	$A = 2$
Class I	v_1	a	$a + c$
Class II	$1 - v_1$	$1 - a$	$1 - a - c$
		$\frac{\text{Var. } A}{1.}$	$\frac{\text{Var. } B}{1.}$
		$\frac{2.}{1.}$	$\frac{2.}{1.}$

Maybe it is hard to believe that two interviews, separated from each other by four weeks, do influence each other that much. Therefore in the next section a more "realistic" example will be given that will be tested against some alternatives.

But before closing this section two more versions of the above model will be shortly discussed, just to show how even on this simple level various models are possible.

The first version is given in Table IV. The main difference between the models of Table III and Table IV is that the "influence" of the first answer is limited to those who scored "1" on variable A . Once again a basic probability of answering "correctly" is assumed. However this basic probability is valid not only during the first interview but also during the second interview for those who scored "2" on variable A . For those scoring "1" on variable A , the probability of answering "1" on variable B will be increased by c . In terms of the example of Table I: if a person has qualified himself during the first interview as a nonvoter, his chances of qualifying himself during the second interview as a voter or a nonvoter are the same as in the first interview. But if a person during the first interview has given the answer "voter", his chances of giving this same answer again are now higher than during the first interview. This model seems particularly appropriate for variables where one alternative is more (socially) desirable than another, or for open questions, where not mentioning things will perhaps not influence the chance of (not) mentioning it a second time. The equations for this model are:

$$p_1^A = av_1 + (1 - a)(1 - v_1) \quad (8)$$

$$p_1^B = av_1(a + c) + (1 - a)v_1a + (1 - a)(1 - v_1)(1 - a + c) + a(1 - v_1)(1 - a) \quad (9)$$

$$p_{11}^{AB} = av_1(a + c) + (1 - a)(1 - v_1)(1 - a + c) \quad (10)$$

From eqns. (8) and (10) one gets the same general solutions for v_1 and a as in eqns. (4) and (7). But c will be estimated differently by:

$$c = \frac{p_1^B - p_1^A}{p_1^A} \quad (11)$$

Applying eqns. (11), (7) and (4) to Table I, one gets:

$$a = .869 \quad c = .075 \quad v_1 = .947$$

In this model the test retest effect is considerably less than in the previous model. There the effect amounted to .188. In this model $(a + c) - a = c = .075$: a person who answered "voter" during the first interview has a .075 higher probability of giving this same answer once more, than a person who answered "nonvoter" during the first interview.

The last model of this section cannot be applied to the example of Table I (or rather, when applied it gives nonsense results). Only the equations and their solutions will be mentioned. As can be seen in Table V, the "influence" of variable A on B is now limited to those who scored "1" on variable A and belong to latent class II: one postulates e.g. that only for those who are not really in the "attractive" class (latent class I) the mentioning of the "attractive" alternative ($A = 1$) will increase their chances of choosing this alternative again. The equations are:

$$p_1^A = av_1 + (1 - a)(1 - v_1) \quad (12)$$

$$p_1^B = av_1a + (1 - a)v_1a + (1 - a)(1 - v_1)(1 - a + c) + a(1 - v_1) \quad (13)$$

$$p_{11}^{AB} = av_1a + (1 - a)(1 - v_1)(1 - a + c) \quad (14)$$

From eqn. (12) one gets for v_1 the same general solution as in eqn. (4).

TABLE V

The parameters of a Model with Direct Effects between Indicators - "Influence" from A to B for Latent Class II

Latent variable	Relative frequencies	Probability of answering			
		$A = 1$		$A = 2$	
		Var. A	Var. B	Var. A	Var. B
		1.	2.	1.	2.
Class I	v_1	a	$1 - a$	a	$1 - a$
Class II	$1 - v_1$	$1 - a$	a	$1 - a + c$	$a - c$
				$1 - a$	a

Eqn. (13) results in:

$$c = \frac{(p_1^B - p_1^A)(2a - 1)}{(p_1^A - a)(a - 1)} \quad (15)$$

Substituting eqns. (4) and (15) into (14) gives:

$$p_{11}^{AB} = a^2 - a + p_1^B, \text{ so}$$

$$a = .50 + \sqrt{.25 - p_1^B + p_{11}^{AB}}$$

4. Latent Probability Models with Test Retest Effects: an elaborate example

To give a more realistic example of the uses of latent probability models with test retest effects data will be used again from the Dutch panel study mentioned in the beginning of Section 2. Through an open question the respondents were asked in November and December, 1970 what kind of problems a new government should first of all pay attention to. The answers were coded from the viewpoint, among others, of house-building.

For the next question a card was given to the respondents on which ten problems were listed, among them house-building. Then the respondent got seven coins and was told that this was some extra money that he, acting as Minister of Finance, should distribute among the ten problems. All this resulted in four variables:

Variable A: answer to open question in November: 1 = house-building mentioned as an important problem; 2 = housebuilding not mentioned.

Variable B: number of coins given in November: 1 = at least two coins to house-building; 2 = less than two coins to house-building.

Variable C: the same as variable A , but in December.

Variable D: the same as variable B , but in December.

Before using these four variables in an analysis of political preferences, three questions should be answered:

Q_1 . Are the open questions and the questions using the coins indicators of the same underlying concept, importance of house-building?

Q_2 . Is this variable (or are these variables) underlying the indicators stable in time?

Q_3 . Does there exist a test retest effect from the first (open) question to the second question using the coins?

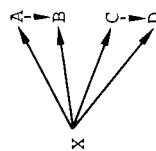
These three questions can be attacked using Table VI within the framework of the latent probability models. At the one extreme there

TABLE VI

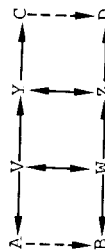
Frequency Distribution of the four Indicators of "Importance of House-Building"

A	B	C	D	Frequency	A	B	C	D	Frequency
1	1	1	1	193	2	1	1	1	58
1	1	1	2	44	2	1	1	2	16
1	1	2	1	26	2	1	2	1	32
1	1	2	2	34	2	1	2	2	48
1	2	1	1	103	2	2	1	1	84
1	2	1	2	77	2	2	1	2	54
1	2	2	1	15	2	2	2	1	60
1	2	2	2	58	2	2	2	2	283
Total									1185

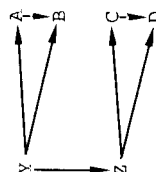
a)



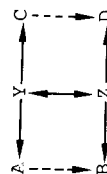
b)



c)



d)



V, W, X, Y, Z : latent variables

A, B, C, D : manifest variables

-----> : possible test
retest effects

Fig. 1. Causal representation of latent probability models.

is the assumption that there exists one latent variable, stable in time, that can explain all the relations between the four indicators A, B, C and D. This is the classical latent class analysis model of Lazarsfeld. Following Goodman (1974b) a causal representation of this model can be found in Fig. 1a. Using Goodman's (1974a, 1974b) algorithm to estimate this model, it turned out that it did not fit the data of Table VI at all ($X^2 = 69.18$, $df. = 6$). Introducing test retest effects did not improve this fit. So one is forced to conclude that the relations between the four indicators cannot be explained by the existence of one stable underlying variable "importance of house-building" (and the presence of test retest effects).

At the other extreme there is the model in which one assumes that each indicator is indicative of only one latent variable; the open questions and the questions using the coins might be measuring different variables, and these underlying variables are not necessarily stable in time. A causal diagram of this model is Fig. 1b. This model has not been investigated further.

Between these two extremes there are the models pictured in Figs. 1c and 1d. In the model of Fig. 1c the assumption is made that the open question and the question using the coins are indicators of the same underlying concept, but that latent changes will take place. Y is the latent variable "importance of house-building" in November, and Z is the latent variable "importance of house-building" in December. Also this model does not fit the data of Table VI ($X^2 = 49.77$, $df. = 4$), and again introducing test retest effects fails to improve this fit.

From the very beginning the model of Fig. 1d was expected to show the best fit with the data. One could suspect that the frames of reference of the open and the "coin question" would be different enough for the respondent to postulate two different underlying variables. And presumably, because of the short period of time between the two interviews, these latent variables would be stable. Also, the occurrence of a test retest effect seemed highly probable: once the respondent has declared house-building is one of the problems the new government should attack first of all, he will try to be consistent, esp. in the presence of the interviewer, and give several coins to this problem on the next question.

Because including or excluding test retest effects made a big difference in terms of the fit of this model, it will be treated here at some length.

First the parameters of this model without test retest effects can be found in Table VII. There are four latent classes, consisting of the combinations of the latent variables Y (the latent "open question") and Z

TABLE VII

The Parameters of the Model, indicated by Fig. 1d, without Test Retest Effects

Y	Z	Latent class	Relative frequencies	Probability of answering															
				Var. A				Var. B				Var. C				Var. D			
				1.		2.		1.		2.		1.		2.		1.		2.	
1	1	I	v_I	a	$1-a$	c	$1-c$	e	$1-e$	g	$1-g$								
1	2	II	v_{II}	a	$1-a$	$1-d$	d	e	$1-e$	$1-h$	h								
2	1	III	v_{III}	$1-b$	b	c	$1-c$	$1-f$	f	g	$1-g$								
2	2	IV	v_{IV}	$1-b$	b	$1-d$	d	$1-f$	f	$1-h$	h								
			$(= 1 - v_I - v_{II} - v_{III} - v_{IV})$																

TABLE IX
The Parameters of the Model, Indicated by Fig. 1d, with Test Retest Effects

Y	Z	Latent class	Relative frequencies	Probabilities of answering
1	1	I	v_I	a
1	2	II	v_{II}	$1-a$
2	1	III	v_{III}	$1-b$
2	2	IV	v_{IV}	$1-b$
			$(=) 1 - v_I - v_{II} - v_{III} - v_{IV}$	
$A = 1$				
$A = 2$				
$C = 1$				
$C = 2$				

or "2" on variable B (D) are different for those who scored differently on variable A (C). The assumption has been made that the mentioning of house-building on the first question increases the probabilities of scoring "1" on variable B (D) by an amount s (t) relative to the probabilities after not mentioning this item on the first question. The model equations are now of the form:

$$p_{1111}^{ABCD} = a(c+s)e(g+t)v_I + a(1-d+s)e(1-h+t)v_{II} \\ + (1-b)(c+s)(1-f)(g+t)v_{III}$$

$$+ (1-b)(1-d+s)(1-f)(1-h+t)v_{IV}$$

$$p_{1112}^{ABCD} = a(c+s)e(1-g-t) + a(1-d+s)e(h-t)v_{II}$$

$$+ (1-b)(c+s)(1-f)(1-g-t)v_{III}$$

$$+ (1-b)(1-d+s)(1-f)(h-t)v_{IV}$$

;

$$p_{2222}^{ABCD} = (1-a)(1-c)(1-e)(1-g)v_I + (1-a)d(1-e)hv_{II} \\ + b(1-c)f(1-g)v_{III} + bdfhv_{IV}$$

Solving these equations by a modified Goodman algorithm [7], the results are encouraging. Granted the use of X^2 is still valid, $X^2 = 4.60$ with two degrees of freedom [8] gives $p = .10$. So it is not necessary to reject the most promising model.

The complete results of Table X may be compared with the results of Table VIII (the analogous model without the test retest effects), but it is simpler to summarize the results in a causal diagram and compare Figs. 2 and 3. Besides the presence of the test retest effects, especially for the second interview, there are two main differences:

(1) The correlation between the latent variables has dropped considerably, from .751 to .445, almost to the level of the manifest correlations mentioned above.

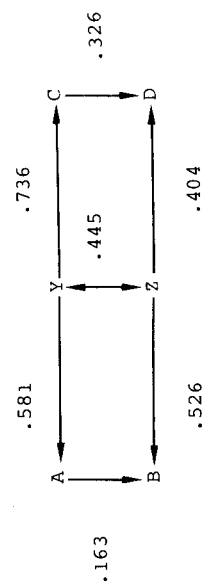


Fig. 3. Causal diagram of the results of Table X.

(2) The effect of Z on D is in the second model substantially less than in the first. Now the conclusion would not be drawn that the second interview gave more reliable results than the first.

Acceptance of this model with test retest effects might once more underline the fact that questions within the same questionnaire or asked within a rather short period of time may have considerable influence upon each other. At the same time, using a model in which these effects are accounted for results in estimates which are cleared of the unwanted side effects.

Notes

- ¹ The most important articles in this respect have appeared in several volumes of the *Sociological Methodology* series. See also Blalock (1971, 1974) and Goldberger and Duncan (1973).
- ² Their first work had already appeared in the fifties: Lazarsfeld (1950) and Wiggins (1955). The best references are now Lazarsfeld and Henry (1968) and Wiggins (1973). See also Harper (1973) and Petersson (1974).
- ³ This research was commissioned by the Catholic Radio's news Programme "Brandpunt" and the Sociological Institute of the Catholic University Tilburg, and carried out by Intomart of Hilversum.
- ⁴ See Wiggins (1973), pp. 124-125 for the general solution of this model.
- ⁵ A number of these models can be found in Wiggins (1973) and Goodman (1974a, 1974b).
- ⁶ In this way Goodman (1974b) makes a link between his work on latent structure models and his contributions to loglinear models.
- ⁷ One way to solve the model of Fig. 1d (with the test retest effects), using Goodman's algorithm is to also consider A , B , C and D as latent variables, and link them with the manifest variables A' , B' , C' and D' respectively with the extreme probability of 1 (and zero) of giving the correct (or incorrect) answer. However in this way it is rather difficult to introduce the necessary constraints on the relative frequencies of the latent classes that are necessary because of the absence of certain arrows and interactions in Fig. 1d. Another approach, used here, follows the logic of Section 3, and estimates the parameters $c + s$, $1 - d + s$, and $g + t$, $1 - h + t$ for $A = 1$ ($C = 1$), separated from c , $1 - d$ (and g , $1 - h$) for $A = 2$ ($C = 2$). The probability s (and t) can be estimated from averaging $(c + s) - c$ and $(1 - d + s) - (1 - d)$ (and $(g + t) - g$ and $(1 - h + t) - (1 - h)$). This estimate of s (and t) can be used to get the final estimates of c and $1 - d$ (and g and $1 - h$). In this particular model the iterative procedure of Goodman did not converge, although the use of different initial estimates of the parameters did give the same minimal X^2 solution. Also with artificial data, this procedure retrieved the true values for the parameters. (In the other models with test retest effects no difficulties were encountered with the convergence of the series.)
- ⁸ The difficulty is whether the procedure described in Note 7 still gives the maximum likelihood estimates, whereby the use of X^2 can be justified. The number of degrees of freedom are set at 2, because this model has two more parameters to estimate than the analogous model without test retest effects which has 4 degrees of freedom.

TABLE X
Estimates of the Model, Indicated by Fig. 1d, with Test Retest Effects

Y	Z	Latent class	Relative frequencies	Probabilities of answering
1	1	I	.410	.719
2	1	II	.151	.281
1	2	III	.124	.719
2	2	IV	.314	.281
$A = 1$				
		Var. B	.713	.187
		Var. C	.187	.713
		Var. D	.287	.813
		Var. A	.813	.287
$A = 2$				
		Var. B	.550	.024
		Var. C	.024	.550
		Var. D	.976	.450
		Var. A	.450	.976
$C = 1$				
		Var. B	.854	.146
		Var. C	.146	.854
		Var. D	.823	.177
		Var. A	.177	.823
$C = 2$				
		Var. B	.497	.503
		Var. C	.503	.497
		Var. D	.581	.419
		Var. A	.419	.581

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WHAT WEIGHT SHOULD WEIGHTS HAVE IN INDIVIDUAL DIFFERENCES SCALING?

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Recently, individual difference scaling has become one of the most active fields of research in psychometrics. Numerous models and algorithms have been proposed but relatively little evidence as to the validity of the produced representations is available so far. To shed some new light on this issue we will reanalyze some data collected by Green and Rao (1972) via PINDIS (Procrustean Individual Difference Scaling). PINDIS is intimately related to the model underlying all presently available individual difference scaling algorithms but differs in important aspects which will allow a deeper insight into validity and interpretability of individual weights generated by them. To make our points clear, we will compare our results with those produced by INDSCAL which is (a) presently the most popular procedure, and (b) also the method of analysis chosen originally by Green and Rao (1972).

The INDSCAL procedure

INDSCAL has been extensively described elsewhere (Carroll and Chang, 1970; Carroll and Wish, 1974a, 1974b) and thus it seems sufficient to outline only its main characteristics here.

The problem in INDSCAL is to scale simultaneously a set of N individual proximity data matrices such that their relationship can be optimally explained by the weighted Euclidean distance model, i.e., by

$$d_{pq}^{(i)} = \left[\sum_{a=1}^m c_a^{(i)} (g_{pa} - g_{qa})^2 \right]^{1/2}, \quad (1)$$